Università di Modena e Reggio Emilia

CASCADAS

Zurich
06/03/2007

Agents and Pervasive Computing Group

Self-organized Knowledge Networks for Situated Autonomic Services

…some ideas on situation-awareness from the CASCADAS project…

Franco Zambonelli

Agents and Pervasive Computing Group

Università di Modena e Reggio Emilia

Università di Modena e Reggio Emilia

CASCADAS

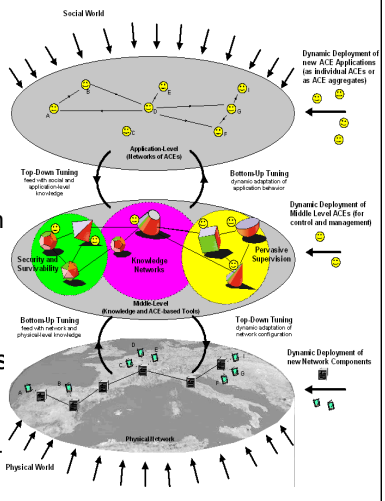
Zurich
06/03/2007

Agents and Pervasive Computing Group

The CASCADAS Integrated Project

Componentware for Autonomic and Situation-aware Dynamically Adaptable Services

- FET "Situated & Autonomic Communication" Initiative, 14 partners, Years 2006-2008
- General objective
  - Identify models and associated tools for a new generation of autonomic and situated communication services
  - Very general concept of "communication services", from network to app level
  - Based on the central abstraction of ACE "autonomic communication element", as th basic building block for complex service networks
  - Eventually leading to the release of an autonomic component framework
- With a specific focus on context-awarenes
  - Services must be situated
  - And there must be tools (i.e., knowledge networks) to support and facilitate context-awareness



Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Towards Situated Services

- Computer-based systems and sensors will be soon embedded in everywhere
  - all our everyday objects
  - all our everyday environments
- Current deployments of pervasive services and sensor networks focus on special purpose systems, e.g.,
  - E.g., environmental monitoring and healthcare
- General purpose approaches are likely to emerge soon
  - Shared infrastructures of sensors, tags, smart-objects, cameras, amd wide availability of Web 2.0 data
  - Generating general purpose data ("facts") about the physical and the social worlds
  - Defining a sort of distributed digital "***world model***"
  - For general-purpose exploitation by situated services (i.e., services for "**browsing the world**")

Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

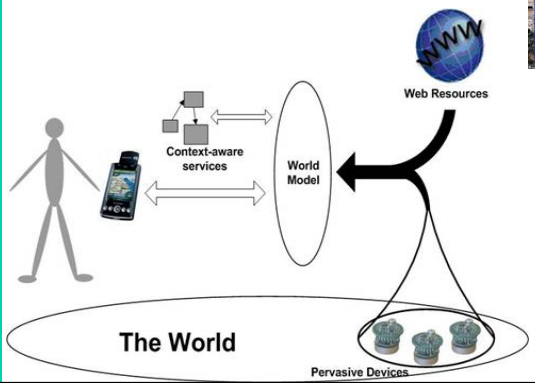
Agents and Pervasive Computing Group

Situated Services Scenario

Exploit the world model to

- Implement services to help us interact with the physical world (e.g., personalized real-time maps)
- Have services coordinate with each other in a context-aware fashion to achieve global goals (e.g., traffic control systems)





PLEASE NOTE: Users and services are themselves part of the world, and thus must be part of the world model.

Data sources and data users are not necessarily distinct entities

Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Autonomic: Where?

- The complexity, openness, and dynamics of the scenario makes it impossible for humans to stay in the control loop
  - Services self-organize and self-adapt to the current world situation
  - Humans can intervene only on limited portions of the scenario
- But…..*is the challenge only with services*?
- Situated services lead to a perspective of mediated interactions
  - Services lives in and access the common "world model" environment
  - Act by sensing it and affect it by acting
  - Components do not need to know each other
  - The environment has its own properties and processes
- But, for making this possible, the world model should be effectively usable by services
  - The challenge is **engineering the environment** (i.e., the world model) other than services



Università di Modena e Reggio Emilia

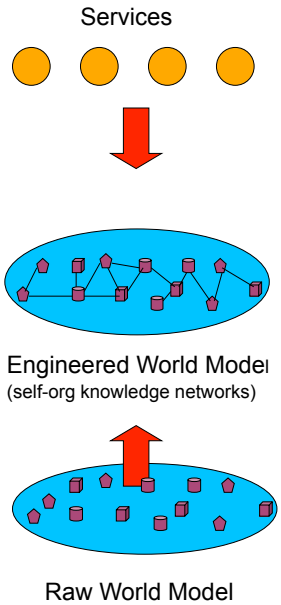
CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Engineering the World Model

- Extreme heterogeneity of data
  - Sensors, cameras, Web 2.0, etc.
  - Variable density (potentially continuum)
- Massive amounts of data produced
  - No way to collect all data at a place
  - Need to aggregate, prune, evaporate, analyse data where it is produced
- Inherent dynamism and decentralization
  - Devices/data come and go at any time
  - No way to control each device due to decentralization
- The necessity arises to **exploit self-organization and autonomic behaviour within the world model**!!!
  - Conceptual shift from self-organizing/ autonomic services to self-organizing/ autonomic data ensembles
  - **Self-organized knowledge networks**



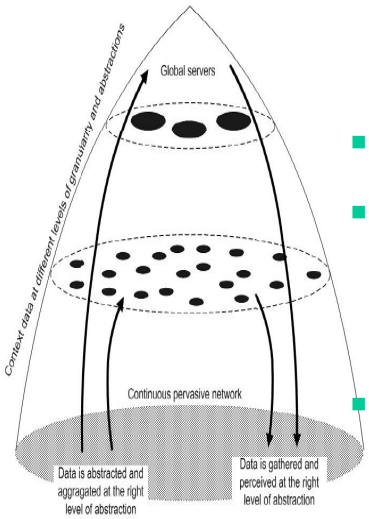
Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

The Knowledge Pyramid



- Data virtually generated in both
  - A sensor network continuum
  - A level of Web 2.0 services
  - At any level in between (e.g., cameras, local servers, etc.)
- To make such data become usable "knowledge" for services it should…
- …flow up the pyramid
  - In aggregated forms, via proper self-organizing algorithms for data aggregation
  - To limit the amount of data (potentially infinite) to be managed by services (which have bounded rationality)
- …flow down the pyramid
  - To self-aggregate data coming from low-level sensors with data coming from higher-level ones (or from the Web)
  - To let services exploit all available data in a uniform way and locally

Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

(Some of the) Challenges

1. Algorithms for self-organized data aggregation (spatial knowledge networks)
2. Uniform and general-purpose data models and algorithms (general-purpose knowledge networks)
3. General-purpose situated service models
4. Security and privacy

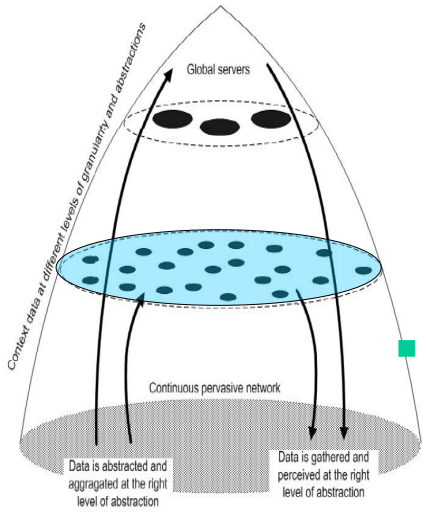
Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Self-organized Data Aggregation



- We need to aggregate data from the continuum
  - Compact representation, manageable by services
  - Detaching from the actual characteristics and density of sensors
  - Without losing relevant information
  - And indeed providing more information of "what's happening"
- A possible idea
  - Identify aggregation regions
  - Enable "per region" views of specific characteristics of the environment
  - As if there were sorts of virtual macro sensors

Università di Modena e Reggio Emilia

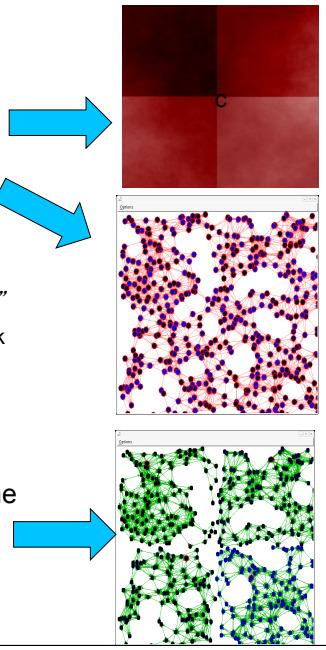
CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Self-organizing Spatial Regions

- Simulated sensors are embedded in an environment characterized by different patterns of sensed data
  At first they are not logically connected with each other
- Gradually they recognize regions
  - A distributed algorithm is continuously running in the network as a sort of *"background noise"*
  - with the goal of partitioning the sensor network into regions characterized by similar patterns (as an overlay of virtual links)
  - Abstracting from the specific density/structure of the sensor network
- Then a partitioning emerges based on the environmental patterns



Università di Modena e Reggio Emilia

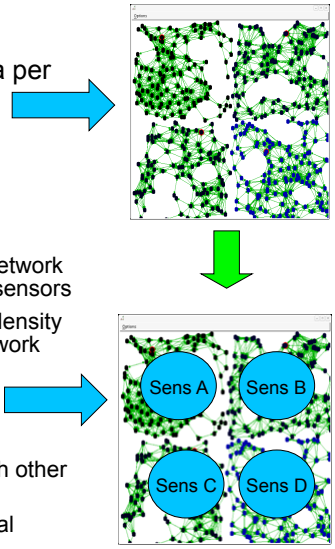
CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Virtual Macro Sensors

- Once Regions are Formed
- Aggregation of data can take place on a per region basis
  - Averaging data over the region
  - Computing regional functions
  - All sensors in region act as a single entity
- Eventually
  - Services can start perceiving the sensor network as composed of a finite number of macro sensors
  - Abstracting from the actual structure and density (potentially continuum) of the physical network
  - Focussing on the real environmental characteristics
- Why we talk of "knowledge networks"??
  - All sensors in a region "network" with each other to act as one
  - Close macro sensors recognize their logical neighbourhood relations



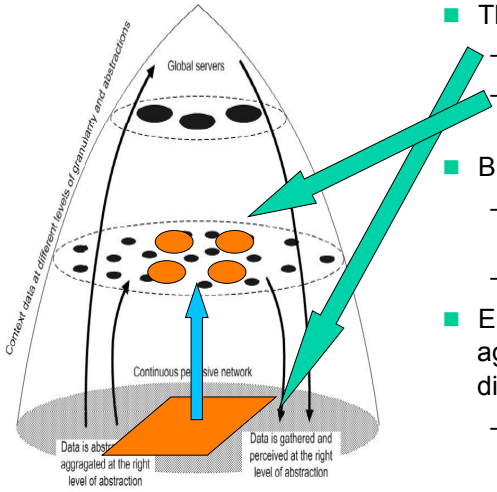
Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

But This is not Enough…



- The presented idea
  - Act on the sensors' continuum
  - To define a manageable virtual macro sensors level
- But we also need to account for
  - Integrating diverse and heterogeneous information
  - Coming from a variety of devices
- Enabling to generalize data aggregation/networking to other dimensions
  - semantics, temporal, application specific
  - other than spatial
- To this end, a ***general knowledge model*** is needed



Università di Modena e Reggio Emilia



Zurich
06/03/2007

Agents and Pervasive Computing Group

Data/Knowledge Models

- How can we provide a uniform representation of data (i.e., facts about the world, that is knowledge about it)
  - Coming from diverse devices (sensors, tags, cameras, and why not Web 2.0 fragments such as geo-tags)
  - Expressing different levels of observations (e.g., a single light sensor vs. a camera)
  - Expressing in a uniform way different redundant perspectives on the same fact (e.g., a WiFi vs. a GPS localization information)
  - Easy to be accesses and manipulated, aggregated (by both services and by within the world model)
- Key guidelines
  - Keep it simple
  - Keep it intuitive
  - Keep it computable
- How do we usually characterize facts about the world?
  - A possible idea: the W4 model



Università di Modena e Reggio Emilia



Zurich
06/03/2007

Agents and Pervasive Computing Group

The W4 Model

- Someone or something (*Who*) does some activity (*What*) in a certain place (*Where*) at a specific time (*When*)
- **Who** is the subject. It is represented by a string with an associated namespace that defines the “kind” of entity that is represented.
  - “person:Gabriella”, “tag:tag#567”, “sensor-region:21”
- **What** is the activity performed. It is represented as a string containing a predicate-complement statement.
  - “read:book”, “work:pervasive computing group”, “read:temperature=23”.
- **Where** is the location to which the context relates.
  - (longitude, latitude),
  - “campus”, “here”
- **When** is the time duration to which the context relates
  - 2006/07/19:09.00am - 2006/07/19:10.00am
  - “now”, “today”, “yesterday”, “before”



Università di Modena e Reggio Emilia



Zurich
06/03/2007

Agents and Pervasive Computing Group

W4 Knowledge: Atoms & Queries

- Gabriella is walking in the campus' park. A service component running on her PDA can periodically create an atom describing her situation.
  - **Who**: user:Gabriella
  - **What**: works:pervasive group
  - **Where**: lonY, latX
  - **When**: now
- Gabriella is walking in the campus, and wants to know if some colleague is near. She will ask (read operation):
  - **Who:** user:*
  - **What:** works:pervasive group
  - **Where**: circle,center(lonY,latX),radius: 500m
  - **When:** now
- Key features
  - Partial knowledge (not fully filled atoms)
  - Context-aware queries ("here", "now")
  - Possibility of generating atoms merging different sources
    - E.g., and RFID atom generated by merging tagID (who), DBMS data (what), GPS data (where), PDA data (when)
  - Device independence



Università di Modena e Reggio Emilia



Zurich
06/03/2007

Agents and Pervasive Computing Group

General-Purpose Self-organized Knowledge Networks

- Knowledge atoms (W4, or whatever) can be related to each other
  - To represent relations between pieces of data and enable navigating related facts of the world
    - Spatial, temporal, or semantic relations
  - To support a better navigation of services in knowledge
  - And more "cognitive" forms of self-organization in services
- Can we exploit self-organization approaches?
  - Self-aggregation of data via semantic/temporal extension of the spatial region approach (cluster and link related data to define multiple and multilevel knowledge views)
  - Have data diffuse (data distribution) and evaporate (data obsolescence) the same as pheromone does in ant-based systems
  - Create knowledge structures that services can perceive as sorts of virtual force fields (context-aware data replication)
  - Matching data patterns and chemical-like reactions (creation of new knowledge)
- A lot of issues to be investigated
  - But ***what service model*** can take advantage of it?



Università di Modena e Reggio Emilia



Zurich
06/03/2007

Agents and Pervasive Computing Group

Service Models

- Situated autonomic services…
  - Location-dependent queries, Social interactions, Real-time understanding of situations, etc.
- …and autonomic communication services
  - Distributed cooperation, Ad-hoc communications, traffic control systems, etc.
- Given the availability of a proper "world model" (we assume that there will be a multiplicity of accessible "spaces" where to store local world models)
  - we have to code specific software components that can somewhat autonomously "query" the world model for
  - Achieving context-awareness and adapt to the current situation
  - Navigating the world model and extract high-level information about situation occurring in the environment
  - Coordinate with each other in a self-organizing way
- So, in the end
  - ***service = autonomic situated communication element = ACE***



Università di Modena e Reggio Emilia



Zurich
06/03/2007

Agents and Pervasive Computing Group

Key Features of Services/ACEs

- We require services ("ACEs") the capability of
  - querying the local world model for extracting info (also as subscriptions to events) → "Knowledge Needed"
  - injecting new data/knowledge atoms and/or new aggregation algorithms in the space → "Knowledge Available"

```
KnowledgeAtom[ ] read(KnowledgeAtom template);
KnowledgeAtom[ ] subscribe(KnowledgeAtom template);
void inject(KnowledgeAtom a);
```

- Also consider that:
  - Services/ACEs themselves (their characteristics and actions) will be represented by some sorts of atoms in the space
  - In the end, in a "smart objects world", objects can generate data and that, by accessing data and by hosting computations, can implemented coordinated services
- After all, this may appear to simply reduce to a "shared blackboard" or "tuple space" interaction model
  - Well, actually, it is more like and "ant-based" self-organization model

Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Browsing the World Services vs. Ants

- Services produce and read knowledge tuples
  - The same as ants release and sense pheromones
  - Typically with very simple local algorithms
- Services indirectly interact via the tuples in the world model
  - Stigmergy + Semantics = Cognitive Stigmergy!
- The world model describe something about the environment
  - "There is another service near here"…the same as pheromones do
- The environment is active
  - Aggregation and self-organization atoms → diffusion and evaporation of pheromones, virtual force fields, chemical reactions, etc.
- We still have to fully unfold these issues, and define a simple and usable service model…
  - And we cannot forget about ***security issues***!

Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Security Issues

(please note I'm not an expert on security!!!)

- Security and trust for services is an issue
- Apparently, there can be an issue also with data:
  - Data affect services behavior at both individual and collective level
  - How to protect data (and the objects/devices from which such data come) from attacks?
- Provocative question: *is this really an issue?*
  - If users/developers have no direct control over the whole system, but only over limited portions of it, how can hackers have?
  - If data is really overwhelming and to most extent intertwined and redundant, how can hackers significantly affect information?
- Examples:
  - Localization: GPS, WiFi, Bluetooth, Mobile Telephony, RFID tags
  - Who can really affect all of these to generate believable localization information?
  - (think also at citation indexes: would you be able to affect both SCOPUS, CiteSeer, DBLP, GoogleScholar, to make people really believe you are a highly-cited author?)

Security vs. Disturbances

- We should try thinking at security in totally different terms
  - Attacks can be perceived as a background noise, a disturbance on the perception of the "world model"
  - Self-organizing services should exhibit dynamics capable of tolerating disturbances
    - As already said, they should be able to deal with partial information, as well as with redundant information
    - They should be able to tolerate inconsistencies
  - Calling for self-organized aggregation algorithms for data redundancy and data consistency against noise
- What about privacy?
  - Do not know, I am tempted to think it is mostly hopeless to strive for it

Università di Modena e Reggio Emilia

CASCADAS

Zurich 06/03/2007

Agents and Pervasive Computing Group

Conclusions

- Future pervasive communications scenarios invites considering
  - A continuum of sensors making available a sort of world model
  - Self-organizing and adaptive services
  - Relying on self-organizing knowledge networks
- Fulfilling the vision is very challenging, calling for
  - General self-organizing algorithms for data/knowledge aggregation
  - Proper general models for representing, accessing, and integrating heterogeneous contextual data
  - Suitable component models exploiting the above for the provisioning of autonomic services
  - Novel, non traditional. perspectives on handling security
- These, and many other issues, are currently under investigation within the CASCADAS project